

UNIVERSAL MULTIMODAL CO-GENERATION WITH CONTINUOUS BITSTREAM DIFFUSION

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ABSTRACT

[Luca will lead the pitch and framing of the paper; to be written last, once results are final.]

1 INTRODUCTION

Continuous generative diffusion models have become an essential component of modern image, video, and audio generation pipelines. The central idea of diffusion modeling is to learn the reverse of a corruption process (i.e., the forward process) that gradually transforms data into a pure noise distribution. While diffusion models have achieved their greatest success on continuous data such as images, recent years have seen growing interest in extending these methods to discrete data modalities, including language and molecular sequences.

In this context, most approaches have focused on discrete forward processes, such as masking or hard token flipping, which progressively replace discrete tokens with other tokens until the original signal is lost. An alternative, first explored in (Vahideh et al., 2023), is to embed the discrete tokens directly into a continuous space and then apply a conventional continuous diffusion process. This approach has seen renewed interest over the past year, with several recent works suggesting that continuous diffusion can match or even surpass the performance of its discrete counterparts. Despite these developments, continuous diffusion approaches are still used primarily for continuous data, whereas discrete diffusion methods, or autoregressive models, remain the dominant choice for discrete sequences. This separation is particularly problematic in multimodal settings, where continuous data, such as molecular geometries, must be co-generated with discrete data, such as molecular identities. Doing so typically requires either integrating substantially different training pipelines within a single model or coupling separate models, both of which can hinder the learning and sharing of genuinely multimodal representations and information. Furthermore, this methodological segregation means that each modality must be tuned and calibrated within its own pipeline, requiring costly, case-by-case optimization and engineering rather than a unified modeling framework or common engineering standard.

In this paper, we introduce a modality-agnostic continuous bitstream diffusion (CoBit) pipeline capable of handling disparate multimodal co-generation tasks without the need for ad hoc calibration, beyond the choice of a digital vocabulary. The multimodal CoBit pipeline converts data from all modalities into sequences of bits, which are then corrupted with continuous Gaussian noise. As in previous work on language (Vahideh et al., 2023), the diffusion process is calibrated using data-dependent entropy profiles, allowing the model to automatically adapt its noise schedule to the underlying data source during both training and inference. As we show, the combination of a shared bitstream representation and entropy-based calibration makes it highly adaptable, enabling strong performance across a broad range of multimodal co-generation tasks with little or no domain-specific tuning.

2 RELATED WORK

[To be written.]

3 METHOD

Our model is the continuous bitstream diffusion model of Batzolis et al. (2026) (CoBit), a text-generation framework, applied *unchanged* to sequences that interleave several modalities. The corruption process, the denoiser parameterisation and its loss, the noise schedule and its estimator, the self-conditioning mechanism, the sampler and the backbone are those of the text-only model, with the same hyperparameters. What is added is small and lives entirely in the data pipeline: (i) a way of writing any set of tokenised modalities into one shared binary vocabulary, so that a single network reads and writes all of them through one interface (§3.2); and (ii) a training recipe under which one set of weights supports unconditional co-generation of all modalities present, conditional generation in every direction between them, and any mixture of the two, with the direction chosen at inference time by deciding which positions are held fixed (§3.3). There are no modality-specific encoders, decoders, heads, embedding tables or losses, and no modality-specific training stages. We recap the base model first, to the extent needed to make these two additions precise.

3.1 CONTINUOUS BITSTREAM DIFFUSION

Bits as the universal representation. CoBit models a sequence of N discrete tokens $t_1, \dots, t_N$ from a vocabulary of size V by writing each token as its $m = \lceil \log_2 V \rceil$ -bit binary index, giving a flat bitstream $\mathbf{x}_0 \in \{0, 1\}^S$ with $S = Nm$. Nothing else about the data is used: the framework does not know whether a token is a word piece, an image patch code, a speech frame or a residue, and it never sees a token identity, only its bits. Once written as bits, the sequence is treated as continuous data. A standard variance-exploding Gaussian diffusion is run on it, $\mathbf{x}_\sigma = \mathbf{x}_0 + \sigma \epsilon$ with $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and $\sigma \in [\sigma_{\min}, \sigma_{\max}] = [0.002, 80]$, taking the data centre to be $1/2$ and $\sigma_{\text{data}} = 1/2$. A denoiser $D_\theta(\mathbf{x}_\sigma, \sigma) \in (0, 1)^S$ predicts the posterior probability that each bit is one and is trained with the EDM-weighted denoising loss (Karras et al., 2022) in probability space,

$$\mathcal{L}(\theta) = \mathbb{E} \left[w(\sigma) \frac{1}{S} \|D_\theta(\mathbf{x}_\sigma, \sigma) - \mathbf{x}_0\|^2 \right], \quad w(\sigma) = \frac{\sigma^2 + \sigma_{\text{data}}^2}{\sigma^2 \sigma_{\text{data}}^2}, \quad (1)$$

which induces the score $\nabla \log p_\sigma(\mathbf{x}_\sigma) \approx (D_\theta - \mathbf{x}_\sigma)/\sigma^2$ used by the sampler. The network is not asked to learn the local Gaussian denoising problem: the analytic posterior logit of an isolated bit under a uniform prior, $(\mathbf{x}_\sigma - \frac{1}{2})/\sigma^2$ (clipped to ± 30), is added to a learned contextual residual, so that the network models only the dependence *between* bits—which is where all modality-specific structure resides. Three components of the base recipe matter for what follows.

Entropy-derived noise schedule. The rate at which information about $\mathbf{x}_0$ is destroyed as σ grows depends on the data, and different modalities—and mixtures of modalities—lose their information at different noise levels. Rather than fixing a proposal over σ by hand, CoBit measures this rate on the data being modelled and allocates both training noise levels and sampling steps according to it. The conditional entropy rate per unit $\log \sigma$ is estimated online from the unweighted per-example denoising error, $\hat{h}(\sigma) \propto e(\sigma)/\sigma^2$ with $e(\sigma) = \frac{1}{S} \|D_\theta - \mathbf{x}_0\|^2$, kept in a FIFO buffer over recent training examples (8×10^5) and binned into 128 log-spaced bins. The training proposal for σ , and through its inverse CDF the sampling grid, is $q(\sigma) \propto g(\sigma; c, n) \hat{h}(\sigma)^{1/2}$, where $g(\sigma; c, n) = \sigma^n / (\sigma^n + c^n)$ ($c = 0.1$, $n = 3$) is a smooth gate removing mass from very small σ . Training starts from the EDM log-normal proposal ($P_{\text{mean}} = -1.2$, $P_{\text{std}} = 1.2$), ramps linearly to q over steps $4\text{--}5 \times 10^4$, and refreshes q every 2×10^3 steps thereafter. Because $\hat{h}$ is measured rather than assumed, the schedule is self-calibrating: the same estimator with the same hyperparameters is used for every modality combination, and it adapts itself to each.

Self-conditioning. Following Chen et al. (2023), the denoiser takes a second input $\hat{\mathbf{x}}_0$, an estimate of the clean bits, concatenated bitwise with $\mathbf{x}_\sigma$. During training, with probability $p_{\text{sc}} = 0.5$ a first no-gradient pass produces $\hat{\mathbf{x}}_0 = D_\theta(\mathbf{x}_\sigma, \sigma, \mathbf{0})$, which is detached and fed to the second, trained pass; otherwise $\hat{\mathbf{x}}_0 = \mathbf{0}$. At sampling time two refresh modes are available. In `carry` mode the probabilities returned at step i are passed as $\hat{\mathbf{x}}_0$ to step $i+1$, so self-conditioning costs no additional network evaluations; in `refined` mode an extra denoiser pass is run at the new state $(\mathbf{x}_{\sigma_{i+1}}, \sigma_{i+1})$ conditioned on the carried estimate, doubling the evaluations per step. We use `carry` throughout unless stated otherwise.

Sampler and backbone. We integrate the probability-flow ODE $d\mathbf{x}/d\sigma = (\mathbf{x} - D_\theta(\mathbf{x}, \sigma))/\sigma$ with a first-order (DDIM/Euler) step (Song et al., 2021) on the entropy-derived grid, optionally with

EDM-style churn (re-noising by a factor $1 + \gamma_i$ before each step; Karras et al., 2022) as the stochastic variant. Integration stops at a terminal $\sigma_{\text{dec}} > \sigma_{\text{min}}$; the final probabilities are thresholded at $1/2$, regrouped into m -bit codes and handed to the appropriate tokenizer’s decoder. The number of steps, the guidance scale (§3.3) and the churn are inference-time choices that require no retraining. The denoiser is the Sequence Diffusion Transformer of Batzolis et al. (2026): the m bits of a position, with their per-bit positional features, form one patch that is linearly projected to one trunk token, so the transformer runs at the token length N rather than the bit length Nm ; the trunk uses RoPE, SwiGLU and adaLN-zero conditioning on σ , and a light skip-connection head combines the trunk output with the raw noisy bits to produce the m per-bit logits of each position. The same backbone is used for every modality combination; only its depth and width vary between checkpoints.

3.2 MULTIMODAL BITSTREAMS

The base model needs nothing from its input except that it be a sequence of fixed-width binary codes. This is what makes it modality-agnostic, and it makes extending it to new modalities a data-preparation step rather than a modelling one. Three ingredients are required.

Modality-specific tokenizers. Each modality is mapped to a run of integer codes by a discrete tokenizer of its own—a byte-pair vocabulary for text, a vector- or lookup-free quantiser for images or audio, a residue alphabet or a structure quantiser for proteins—and back by the matching decoder. The tokenizers are the *only* modality-specific components in the system. They are frozen, they are not trained or fine-tuned jointly with the diffusion model, and they sit outside it: the diffusion model never sees an embedding, a spectrogram or a pixel, only bits. Any tokenizer that yields fixed-width codes can be used; quantisers whose codebook is the full set of sign patterns of a latent vector (lookup-free or finite-scalar quantisation; Yu et al., 2024; Mentzer et al., 2024) pair especially naturally with bitstream diffusion, since every bit string is then a valid code and the bits are the quantiser’s own latent coordinates rather than an arbitrary index. The tokenizers used for each modality pair are stated in the corresponding experimental subsection (§4).

One shared code space. The per-modality vocabularies are placed in disjoint ranges of a single integer code space, together with a small reserved block of control codes (padding and segment markers), and m is chosen as the bit width of that union—in practice by offsetting the vocabularies or, equivalently, by prepending one or more namespace bits that select the modality. The m bits of a position therefore determine both the modality and the content of a token, every position is a “token of any modality”, and the patch size of the backbone is m for all of them. Consequently there are no modality-specific embedding tables, encoders, heads or losses, and the shared Bernoulli output head costs $O(m) = O(\log V)$ per position regardless of how many vocabularies are merged. The one modality-aware parameter we add is a learned segment embedding: a zero-initialised vector per segment type (control, text, image, audio, ...) added to the per-bit trunk input, which tells the network which segment a bit belongs to without introducing any per-modality computation.

A fixed sequence template. A training example is laid out as a fixed-length template of segments, one per modality, each of a fixed token budget (truncated or padded), separated by structural markers (start/end of sequence and of each segment). Markers are ordinary codes in the reserved block; they are always presented clean, at training and at sampling, and are never denoised. Two-dimensional or higher-dimensional token grids are flattened along a locality-preserving (Hilbert) curve so that 1-D adjacency reflects spatial adjacency. Modalities that are aligned position by position (for instance a residue identity and a structure token describing the same site) may also share one position by concatenating their codes into a single m -bit word; this too is only a layout choice. *[Confirm against the final protein codec in §4.5: the current stub describes an 18-bit position formed by a 5-bit residue code and a 13-bit DPLM-2 structure code.]* The transformer sees only the flat bitstream; the same code path decodes each segment through its own tokenizer’s decoder.

What is, and is not, modality-specific. Across every experiment in this paper the corruption process, the σ range and σ_{data} , the loss and its weighting, the entropy-schedule estimator and all of its hyperparameters, self-conditioning, the sampler family and the backbone architecture are identical. *[Verify against the final audio and protein training configurations before submission; this sentence is the paper’s central claim and should be checked field by field.]* What changes between modality pairs is confined to the data pipeline: the tokenizers and their decoders, the segment layout (which modalities, in which order, with what token budgets), and the list of training regimes of §3.3.

The only quantities chosen *per direction* at inference are the sampler settings (step count, guidance scale, churn), which are free hyperparameters of any diffusion sampler and involve no retraining and no architectural difference.

3.3 GENERATIVE DIRECTIONS

A sequence built as above is partitioned into segments $S_1, \dots, S_K$ (one per modality, plus markers). Any generation task over these modalities is a choice of a *conditioning set* $\mathcal{C}$ of segments that are given and a *target set* $\mathcal{T}$ that must be produced: $\mathcal{C} = \emptyset$ is unconditional joint co-generation of everything; conditioning on a text segment gives text-to-image or text-to-speech; conditioning on an image or a speech segment gives captioning or transcription; conditioning on a prefix of a segment gives continuation; and so on. CoBit implements every such task with the same network and the same weights through a single mechanism, which we call *partial denoising*: the positions in $\mathcal{C}$ are held at their clean values x_0 throughout, and diffusion acts only on the positions in $\mathcal{T}$. Because the model is a bidirectional denoiser over the whole sequence, a clean segment is simply part of the input it conditions on; no task token, task-specific head or encoder is involved.

Training. For each training example we draw a task from a small list of *regimes*—conditioning sets with sampling probabilities—and construct the noisy input accordingly. Let $M \in \{0, 1\}^S$ be the bit mask of the conditioned positions (the segments in $\mathcal{C}$, always including the markers). The input to the denoiser is $x_\sigma = M \odot x_0 + (1 - M) \odot (x_0 + \sigma\epsilon)$, i.e. the conditioning segments are clamped clean rather than noised, and the loss equation 1 is averaged only over the denoised positions $1 - M$. When self-conditioning is active the clean conditioning bits are also written into $\hat{x}_0$ on the masked positions, so the network never has to predict what it is given. To enable classifier-free guidance (Ho & Salimans, 2021), with probability $p_{\text{uncond}} = 0.1$ the conditioning segments (but never the markers) are replaced by the null value $1/2$ —the data centre and the point of zero information for a bit under Gaussian corruption—so that the network also learns the unconditional denoiser for each target set. The regime list is the whole specification of which directions a model supports: adding a direction for a given modality set means adding an entry to this list, without touching the model. Each experimental subsection states the regimes used. Examples whose conditioning was dropped are excluded from the entropy-rate buffer of §3.1, so the schedule reflects the conditional denoising problems the model is actually asked to solve.

Sampling. At inference the task fixes M and the clean values on the conditioned positions; the sampler of §3.1 initialises the target positions from noise, clamps the conditioned positions to their clean bits before every denoiser call, zeroes the ODE velocity on those positions after every call (so they never move) and clamps them once more in the returned probabilities. Guidance uses the standard two-branch estimate in probability space: a conditional branch sees the clean conditioning, an unconditional branch sees the null value $1/2$ on the same positions (markers protected), the two are evaluated as one batch of size $2B$, and the guided estimate $D_g = D_u + w(D_c - D_u)$ replaces D_θ in the score, where $w = 1$ recovers the plain conditional model and $w > 1$ sharpens it. In *carry* mode each branch carries its own self-conditioning estimate. Joint co-generation ($\mathcal{C} = \emptyset$) uses a single branch, so its cost per step is one network evaluation while every guided conditional direction costs two; we report the number of function evaluations (NFE) accordingly. Since the direction is decided entirely by M , the remaining sampler choices—step count, guidance scale, churn—can be set per direction on the same weights, and nothing in the framework requires them to agree across directions. The same weights therefore act as an unconditional co-generator, as a family of conditional generators in every direction seen in training, and as a model that fills in an arbitrary subset of positions given the rest.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

[Shared setup common to all modalities.]

4.2 IMAGE GENERATION (CLASS-CONDITIONAL IMAGENET)

[Georgios to write: image-only results and comparisons.]

Direction	n	Steps (NFE)	FID↓	CMMD↓	CLIP↑	CIDEr↑
<i>CoBit-CC3M — one set of weights (462M, $f16$, 256 image tokens)</i>						
text → image	30,000	10 (20)	18.48	18.41	27.73	—
image → text (CLIP)	5,000	256 (512)	—	—	24.83	—
image → text (captioning)	5,000	256 (512)	—	—	—	0.136
joint, 10 steps	30,000	10 (10)	50.19	41.53	23.10 [†]	—
joint, 256 steps	30,000	256 (256)	61.95	46.80	25.67 [†]	—
<i>CoBit-CC12M — one set of weights (633M, $f8$, 1,024 image tokens)</i>						
text → image	30,000	16 (32)	16.59	17.76	26.51	—
image → text (CLIP)	5,000	256 (512)	—	—	22.98	—
image → text (captioning)	5,000	256 (512)	—	—	—	0.057
joint, 20 steps	5,000 [‡]	20 (20)	56.18	37.60	23.19 [†]	—
joint, 256 steps	5,000 [‡]	256 (256)	64.47	40.33	24.01 [†]	—
<i>rFID floor of the tokenizer, $f16 / f8$</i>	30,000		1.59 / 0.73	—	—	—

Table 1: Zero-shot MS-COCO, all four generative directions from a single set of weights per model. Each conditional row is quoted at the step count that is best for that direction (*Sampler* below); joint (unconditional) generation has no single optimum, so both ends of its trade-off are shown. CLIP is ViT-B/32; per-direction guidance and churn are listed in Appendix A.1, and ViT-L/14 scores, SWD and precision/recall in Appendix Table 6. [†]Pair coherence $\cos(\text{generated image}, \text{generated caption})$, not comparable to the conditional CLIP scores; FID of the joint rows is of the generated image half. [‡]FID is strongly biased in n (+5.3 at $n=5000$ versus 30,000), so these rows are not FID-comparable to CC3M’s (CC3M at $n=5000$: 56.06 / 67.47 at 10 / 256 steps).

4.3 IMAGE AND TEXT (CC3M, CC12M)

Setup. We train CoBit-CC3M (462M parameters, 24×1024) on Conceptual Captions 3M (Sharma et al., 2018) ($\approx 2.9\text{M}$ pairs, 10^6 steps) and CoBit-CC12M (633M, 26×1152) on Conceptual 12M (Changpinyo et al., 2021) ($\approx 11.7\text{M}$ pairs, 1.744×10^6 steps, 87% of its planned budget). Images are tokenized by a frozen Open-MAGVIT2 lookup-free quantiser (Luo et al., 2024; Yu et al., 2024) into 18-bit codes, $16 \times 16 = 256$ tokens ($f16$) for CC3M and $32 \times 32 = 1,024$ ($f8$) for CC12M; text by the $\text{o}200\text{k}$ byte-pair vocabulary (caption budgets of 48 and 96 tokens). A 19th namespace bit separates image from text and control codes, so every position is one 19-bit token (§3.2); training regimes are 0.35 text-conditioned, 0.35 image-conditioned, 0.30 joint. Neither model sees MS-COCO (Lin et al., 2014): all numbers are zero-shot on COCO val2014 under one fixed protocol (FID-30K, CMMD (Jayasumana et al., 2024), CLIP ViT-B/32 (Radford et al., 2021); captioning on the Karpathy test split; Appendix A.1). Deliberately absent are text-only pre-training, tokenizer surgery, auxiliary alignment losses and staged curricula. All are known to help and we expect them to help here; we omit them so that the result measures what the unmodified framework transfers, and leave them to future work.

Results. Table 1 reports all four directions for both models, each from a single set of weights. CC3M reaches zero-shot FID-30K **18.48** at NFE 20; its tokenizer’s own reconstruction floor on the same 30,000 images is rFID 1.59, so the codec accounts for under 2 of those points and the gap to larger systems lies in the model, not in the token space. Joint co-generation is much harder than either conditional direction (FID 50.19 at 10 steps), as with no prompt the model samples its own CC3M-like training distribution, yet the co-generated image and caption agree with each other (pair coherence 25.67 at 256 steps) about as well as a conditional caption agrees with its image (24.83).

Comparison to prior work. Table 2 places the result among published zero-shot MS-COCO numbers. Exact comparability is not attainable in this setting and we do not claim it; the table is read for where the model sits, not for rank. Against the one system with the same pre-training corpus and protocol, LAFITE, FID improves by 8.46 without its frozen CLIP encoder. Among unified image+text models we sit between CoDI (400M pairs) and SEED-X (17B parameters), and ahead of the DALL-E and CogView specialists at 12B and 4B parameters. Every system with a lower FID than ours relies on one or more of a text-pretrained backbone or pretrained text encoder, a pretrained CLIP or VAE, and at least an order of magnitude more paired data—precisely the ingredients withheld here. [Verify this sentence row by row against the cited papers before submission.] On GenEval,

Model	Params	Pre-train pairs	FID-30K↓	GenEval↑
<i>CoBit (ours; from scratch, no pretrained components)</i>				
CoBit-CC3M	462M	2.9M	18.48	0.249*
CoBit-CC12M	633M	12M	16.59	—
<i>Same pre-training corpus (CC3M), zero-shot COCO</i>				
LAFITE (Zhou et al., 2022)	75M (+CLIP)	2.9M	26.94	—
<i>Unified image+text models</i>				
CoDI (Tang et al., 2023)	—	400M	22.26	0.31
SEED-X (Ge et al., 2024)	17B	—	14.99	0.49
LWM (Liu et al., 2024)	7B	—	12.68	0.47
UniDiffuser (Bao et al., 2023)	~1B	LAION-5B (subsets)	9.71	—
Show-o (Xie et al., 2024)	1.3B	35M	9.24	0.68
<i>Text-to-image specialists</i>				
DALL·E (Ramesh et al., 2021)	12B	250M	27.50	—
CogView (Ding et al., 2021)	4B	30M	27.10	—
minDALL·E	—	—	—	0.23
LlamaGen	0.8B	—	—	0.32
LDM (Rombach et al., 2022)	1.4B	400M	12.64	0.37
DALL·E 2 (Ramesh et al., 2022)	6.5B	650M	10.39	0.52
SD v1.5 (Rombach et al., 2022)	0.9B	2000M	9.62	0.43
PixArt- α (Chen et al., 2024)	0.6B	25M	7.32	0.48
Imagen (Saharia et al., 2022)	3B	860M	7.27	—

Table 2: Zero-shot MS-COCO FID-30K and GenEval overall score against published systems. Prior figures are as printed in the cited papers and in the compilation tables of Xie et al. (2024) (Tables 2–3), Bao et al. (2023) (Table 1), Zhou et al. (2022) (Table 2) and Ghosh et al. (2023) (Table 2); parameter and pair counts as printed by Xie et al. (2024) where available. Rows are not strictly comparable (see text): corpora, backbones, pretrained components and FID pipelines all differ, and the same nominal metric varies by about one FID point between sources (Stable Diffusion is 8.59 in one compilation and 9.62 in another). Improved VQ-Diffusion’s 8.44 (Tang et al., 2022) is omitted because that model is trained on COCO. *Measured at 256 sampling steps, $25\times$ past the image optimum; a lower bound.

0.249 sits above minDALL·E and below CoDI and LlamaGen, far below the LLM-backed unified models. For captioning no zero-shot COCO baseline has been published; COCO-trained specialists reach CIDEr 1.02–1.40 against our 0.136 (*Limitations* below). Taken together, a 462M-parameter model trained from scratch on 2.9M alt-text pairs, with no modality-specific machinery, generates images in the range of unified systems one to two orders of magnitude larger and captions and co-generates from the same weights. That is the claim: capable transfer, not a leaderboard entry.

Sampler: the two modalities want step counts $25\times$ apart. Sweeping the step count from 4 to 512 on the same weights (Appendix A.3), text→image FID and CMMD minimise at 10 steps for CC3M (16–20 for CC12M) and degrade monotonically beyond; at $n=30,000$ the conventional 256 steps costs 4.06 FID for $25.6\times$ the compute. Captioning inverts this: CIDEr is 0.013 at 10 steps, peaks at 0.136 at 256 and falls to 0.103 at 512 (image→text CLIP behaves alike). The joint direction shows both effects inside single samples: every step added improves the text half and degrades the image half (10, 32, 64 steps at $n=5000$: pair-CLIP 23.16, 24.71, 25.17; FID 56.06, 59.14, 62.66), hence the two joint rows of Table 1. This is not a modification of the framework: the step count, like the per-direction churn and guidance (Appendix A.3), is an inference-time hyperparameter of any diffusion sampler, set per direction on the same weights without retraining—and such sampler settings are the only thing that had to be adapted per modality.

Scaling the corpus. Moving to CC12M ($4\times$ the pairs, $1.37\times$ the parameters, $\approx 8.7\times$ the compute, a better codec) improves images and degrades text: text→image FID falls $18.48 \rightarrow 16.59$, a margin that survives subtracting each tokenizer’s own rFID floor (16.89 versus 15.87; CMMD, 18.41 $\rightarrow$ 17.76, has no such correction and retains an unquantified codec advantage), while at a matched 256 steps captioning CIDEr falls $0.136 \rightarrow 0.057$ and image→text CLIP $24.83 \rightarrow 22.98$. This split is what noisier CC12M alt-text predicts and uniform undertraining does not, but that is an inference

from a pattern across metrics, not a measurement; the decisive checkpoint-progression test has not been run (Appendix A.4).

Limitations. Compositional prompt following is near the floor: CC3M scores 0.249 on GenEval (Ghosh et al., 2023) (single object 0.80, colours 0.46, position 0.04, attribute binding 0.03; Appendix A.5). This was measured at 256 steps, $25\times$ past the image optimum, so it is a lower bound; CC12M has not been evaluated on GenEval. Captioning is weak in absolute terms (CIDEr 0.136, roughly $7.5\times$ below the COCO-trained L-Verse (Kim et al., 2022) at 1.022; no zero-shot COCO captioning baseline has been published), and the outputs are fluent CC3M-style alt-text scored against COCO’s terse references (RefCLIPScore (Hessel et al., 2021) 0.661; Appendix A.6). Both limitations are consistent with CC3M alt-text rarely specifying relations, attribute bindings or COCO-style descriptions, and both are exactly what the omitted ingredients—text pre-training, curated or re-captioned data, alignment losses—are expected to buy. That is the point of the experiment: it shows what unmodified transfer does and does not provide.

4.4 AUDIO AND TEXT

[George to write: audio+text results and comparisons.]

We tokenize speech with StableCodec (Parker et al., 2025), which performs FSQ (Mentzer et al., 2024) on the latent space of a transformer autoencoder. The resulting sequences represent speech at a fixed token rate of 25 tokens per second from a vocabulary of size 46,656. While RVQ (Zeghidour et al., 2021) is the predominant choice for speech tokenization, it would be problematic in the diffusion setup: flattening the codebooks like Llama-Mimi (Sugiura et al., 2025) would introduce strict positional dependencies and major computational overhead (bidirectional attention scales quadratically with sequence length); treating the Cartesian product of codebooks as a single vocabulary would neglect their conditional structure. FSQ-based tokenization is hence more suitable for bitstream diffusion, but StableCodec does not explicitly model speaker identity. To control voice characteristics, we augment the input sequence with speaker tokens. BiCodec-global (Wang et al., 2025) performs FSQ on the outputs of an ECAPA-TDNN (Desplanques et al., 2020) speaker verification model with a vocabulary of 4,096 entries to produce a fixed-length sequence, independent of the input duration. Speaker tokens are not used during waveform decoding; they are included purely as a conditioning signal for generation. Combining these with the text vocabulary used for image-text experiments (4.3) results in a vocabulary of 250,772 tokens (with separate pad tokens for text and speech) and a sequence of 632 tokens in total: 100 text tokens, 32 speaker tokens, 500 speech tokens.

Four speech-native tasks are considered: unconditional co-generation, speech continuation, zero-shot text-to-speech (TTS) and automatic speech recognition (ASR). In co-generation, the entire sequence is generated from scratch; in speech continuation, speaker and speech tokens are extracted from the start of the audio (typically the first 3 seconds). For ASR, the complete speaker and speech sequences are used as input; for TTS, text tokens and BiCodec-global tokens from a held-out recording of the target speaker are provided. Additional tasks that fit the text-speaker-speech structure can be supported simply by modifying the training recipe (3.3). The model is trained on 46k hours of LibriTTS (Zen et al., 2019) and MLS-English (Pratap et al., 2020) data, divided in 11.8 million sequences that contain single-speaker, reading-style paired text-audio samples derived from the LibriVox audiobook corpus. Equal training effort is devoted to each of the four tasks. Evaluation is conducted on the LibriSpeech-PC (Meister et al., 2023) test partitions and the SALMon (Maimon et al., 2025) benchmark. The final sampler configurations (stochasticity and guidance scale) for each conditional task have been selected based on modality alignment (WER) on 256 samples from the LibriTTS dev-clean set.

Autoregressive Speech Language Models (SLMs) are typically not evaluated on paired unconditional generation. They either cannot produce both modalities simultaneously or cannot operate without any input. Instead, we compare CoBit’s co-generation abilities to an autoregressive system that generates one modality at a time. Results are provided in Table 14, and more details about the cascaded pipeline can be found in Appendix B.1.3. CoBit produces strongly aligned text and audio with very few conflicts at the character and word levels, matching the text-LM+TTS combination of the cascade. In terms of perplexity under an external LM, the content remains linguistically plausible although scoring worse than the text-LM-based generation. The disparity between text and speech

perplexities for CoBit can largely be attributed to the `Whisper-large` model used to transcribe the audio, which normalizes its outputs into well-structured sentences. Notably, the model generates natural-sounding speech even when sampling the speaker-conditioning tokens from scratch. While the cascade system achieves a higher UTMOS score, it does so by conditioning the speech decoder on BiCodec-global tokens, strictly forcing voice and prosody preservation throughout the audio.

Speech continuation is the main generative evaluation axis for autoregressive SLMs. Given a short audio prompt, models are tasked with producing continuations that match both the acoustic setting and the semantic content. Table 15 reports evaluation results on LibriSpeech-PC test-clean. The model consistently achieves the second best score in all aspects except LLM-as-a-Judge, ranking only behind TASLM-Embed (Tseng et al., 2025). CoBit’s continuations are linguistically plausible, achieving very low perplexity and comfortably outperforming models with over a billion parameters. While constructing well-formed sentences, most SLMs are incapable of producing relevant continuations to the prompt’s content. The same failure is observed for CoBit, which performs moderately worse than almost all autoregressive models that operate on fixed-rate speech tokenization, but differ in scale and text pre-training (Hassid et al., 2023). Even warm initialization, however, appears to be insufficient for semantic extrapolation; thematically accurate speech is only achieved by TASLM-Embed, which uses text-aligned speech representations that close the semantic gap between text and speech modelling. Reinforcing the co-generation results, CoBit generates decisively more natural audio than all models that do not condition the decoder on speaker-related information. It is also able to match distributional characteristics of the ground-truth continuations in terms of frame-level details and expressiveness better than all such models, outperforming them in FSD metrics by a wide margin. This can be credited to classifier-free guidance, which steers generation towards the mode of the prompt-conditional distribution.

Generating speech requires capturing multiple, often unrelated aspects that form the linguistic and acoustic content, such as syntax or background noise. For autoregressive models, such properties are evaluated in isolation using likelihood-based protocols (Hassid et al., 2023; Maimon et al., 2025; Nguyen et al., 2020): the SLM is provided with the true sample and an alternative version where the property of choice is modified, and is tasked with assigning a higher likelihood to the positive example. However, the likelihood estimation capabilities of diffusion models are not representative of their generative capacity (Theis et al., 2015). Instead, we adopt the recently proposed generation-based version of the SALMON benchmark (Hsu et al., 2026) to assess CoBit’s ability to capture acoustic details (see Appendix B.1.2 for more details). Table 16 presents the per-task scores. On average, CoBit outperforms all other text-speech hybrids and the Flow-SLM-S (Chou et al., 2025) variant, scoring below larger speech-only models with over 1B parameters. For voice-related tasks (sentiment, speaker, gender), CoBit is on par with the strongest baselines, with the aid of the BiCodec-global features that capture such characteristics. In noise-related tasks (background and room), the model sits in between the two clusters of baselines: it does better than the rest of the multimodal SLMs that focus heavily on speech semantics, yet it cannot match the speech-only baselines that use the high-bandwidth Mimi tokenizer (Défossez et al., 2024), which is trained on a larger and more diverse corpus than StableCodec and can thus capture more fine-grained details in the audio signal that are not necessarily related to the actual speech.

Partial denoising enables straightforward text-to-speech and speech-to-text transfer for CoBit; this is not the case for other SLMs, which require task-specific fine-tuning or engineering adjustments like few-shot prompting and time-delayed streaming generation (see Appendix B.1.3 for an overview). TTS and ASR performance is outlined in Table 17. As already observed for co-generation, bit-stream diffusion produced very few errors in text and speech alignment. For TTS in particular, CoBit is competitive with SLMs with up to 7B parameters: it surpasses SpiritLM (Nguyen et al., 2025) on CER (4.1% compared to 6.7%) and approaches Moshi (Défossez et al., 2024) on WER (5.3% compared to 4.7%). While the generated speech remains natural, it cannot exactly reconstruct the required speaker characteristics: the BiCodec-global conditioning steers the speech tokens towards the provided voice, but their absence from the decoder prevents an exact speaker match. The contrast in SpkSim with GPA-v1.5 (Cai et al., 2026), which has a similar scale at 505M parameters and conditions its vocoder to the same BiCodec-global tokens, further verifies this finding. Despite its TTS success, CoBit cannot transcribe speech as accurately as text-initialized systems. This is expected when looking at the design of ASR systems: they typically recognize speech at the character or phoneme level using self-supervised representations, and then employ a text LM to decode their output into well-structured sentences (Radford et al., 2023). For SLMs, the textu-

Direction	Evaluation	$n \times k$	Recovery $\uparrow$	TM/scTM $\uparrow$	IDDT/pLDDT $\uparrow$	RMSD/scRMSD $\downarrow$	Local design $\uparrow$
Inverse	CATH 4.3 val	128 $\times$ 4	44.10 (45.20)	.739	61.80	6.104	88.28%
Forward	CAMEO 2022	183 $\times$ 4	—	.554 (.585)	.541 (.564)	11.316 (9.978)	—
Joint	lengths 100–500	80 $\times$ 1	28.95	.899	72.73	3.659	100.0%

Table 3: Headline results from one checkpoint and one sampler setting. Recovery is amino-acid recovery for inverse folding and sequence recovery for co-generation. TM/RMSD measure forward target agreement; scTM/scRMSD compare a generated structure with an independent refold of its sequence. Parentheses are best-of-four; other entries are means. Local designability uses the evaluation suite’s original gate, scRMSD $< 2 \text{ \AA}$ or scTM > 0.5 ; it must not be confused with the stricter per-metric thresholds in Table 4.

ally pre-trained backbone implicitly plays a similar role, leading to more coherent transcriptions, as previously demonstrated (Maiti et al., 2024).

The multimodal CoBit framework consistently showcases strong performance across all text-audio tasks, balancing modality alignment, linguistic fluency and acoustic naturalness. Notably, it is also constitutes the first fully non-autoregressive SLM, as all other architectures generate autoregressively and strictly non-autoregressive systems are restricted to ASR-TTS hybrids. Its core weaknesses (ASR, thematic preservation and exact voice reconstruction) can be tied to two design choices that baselines consistently exploit: initialization from a text pre-trained transformer with the substitution of its input and output layers and explicit speaker conditioning on the decoder. CoBit is instead trained from scratch on paired data, and waveforms are generated exclusively from StableCodec tokens. These are deliberate choices: the aim of this work is to show that continuous bit-stream diffusion has strong multimodal learning capabilities without modality-specific adjustments. Both modifications are straightforward in principle: warm-initializing CoBit from a pre-trained text checkpoint mainly required adapting the components that are tied to the sequence length and the vocabulary bit-width; speaker conditioning for the decoder is simply a matter of fine-tuning on the joint distribution of BiCodec-global and StableCodec tokens. We expect both changes to close the remaining performance gap and leave their implementation to a dedicated line of work on text-audio CoBit.

4.5 PROTEIN SEQUENCE AND STRUCTURE

Setup. We train CoBit-Protein (396M parameters, 20×1024) from scratch on paired PDB and AFDB–SwissProt sequence–backbone data, with 20% UniRef50 sequence replay. Each residue is represented by one 18-bit token: 5 amino-acid bits and 13 bits from a frozen LFQ structure tokenizer. The same continuous-bit denoiser performs inverse folding, forward folding, and unconditional sequence–structure co-generation; only the observed bits change. We freeze the checkpoint at 245,000 steps and evaluate four samples per target on CATH 4.3 validation (128 inverse-folding backbones) and frozen CAMEO 2022 (183 forward-folding sequences), together with 80 co-generated pairs (16 at each length 100–500). From a $6 \times 5 \times 4$ sweep over classifier-free guidance (CFG), entropy/Karras mixing α , and sampling steps, we select one setting using only the pre-declared pure-entropic inverse-folding arms: CFG 5, $\alpha = 0$, and 50 steps. Forward and co-generation results do not enter selection (CFG is inactive for unconditional co-generation).

Results. One set of weights supports all three directions: inverse folding reaches 44.10% mean recovery, forward folding reaches .585 best-of-four TM-score, and joint generation reaches .899 mean scTM and 3.659 $\text{\AA}$ mean scRMSD (Table 3). Under the stricter thresholds adopted by SimpleDesign, CoBit obtains .39/.69 ($D_{\text{RMSD}}/D_{\text{TM}}$), between SimpleDesign’s high-fidelity $\gamma = .3$ operating point (.53/.74) and its more stochastic $\gamma = .7$ point (.36/.55). CoBit also exceeds the thresholded DPLM-2 and ESM3 rates in that compilation, although DPLM-2 has higher mean scTM and MultiFlow remains strongest overall (Table 4). These comparisons are context rather than a leaderboard: even nominally identical self-consistency metrics depend on the refolding judge and sampling protocol.

The comparison with SimpleDesign sharpens the architectural result. SimpleDesign couples discrete amino acids to continuous C_α coordinates without a structure tokenizer (Lu et al., 2026); CoBit instead compresses both modalities into bits, then uses exactly the same denoiser and clamping

Model	Params	n	Mean scTM $\uparrow$	Mean scRMSD $\downarrow$	$D_{\text{RMSD}}\uparrow$	$D_{\text{TM}}\uparrow$
CoBit-Protein	396M	80	.899	3.659	.39	.69
DPLM-2 (Wang et al., 2024)	650M	100	.925	3.899	.30	.46
ESM3 Open (seq $\rightarrow$ str) (Hayes et al., 2025)	1.4B	100	.624	24.180	.09	.11
MultiFlow (Campbell et al., 2024)	—	100	.930	3.208	.76	.80
SimpleDesign ($\gamma = .3$) (Lu et al., 2026)	—	100	—	—	.53	.74
SimpleDesign ($\gamma = .7$) (Lu et al., 2026)	—	100	—	—	.36	.55

Table 4: Protocol-aware co-generation context for lengths 100–500. Following Lu et al. (2026), D_{RMSD} is the fraction with scRMSD ≤ 2 Å and D_{TM} the fraction with scTM $\geq .9$; CoBit’s fractions are recomputed from its 80 per-sample outcomes. Published baseline fractions and $n = 100$ are from their Table 1; mean columns are the intersection previously reported by the corresponding methods. Different training sets, refolding pipelines, sample counts, and baseline implementations preclude a strict ranking.

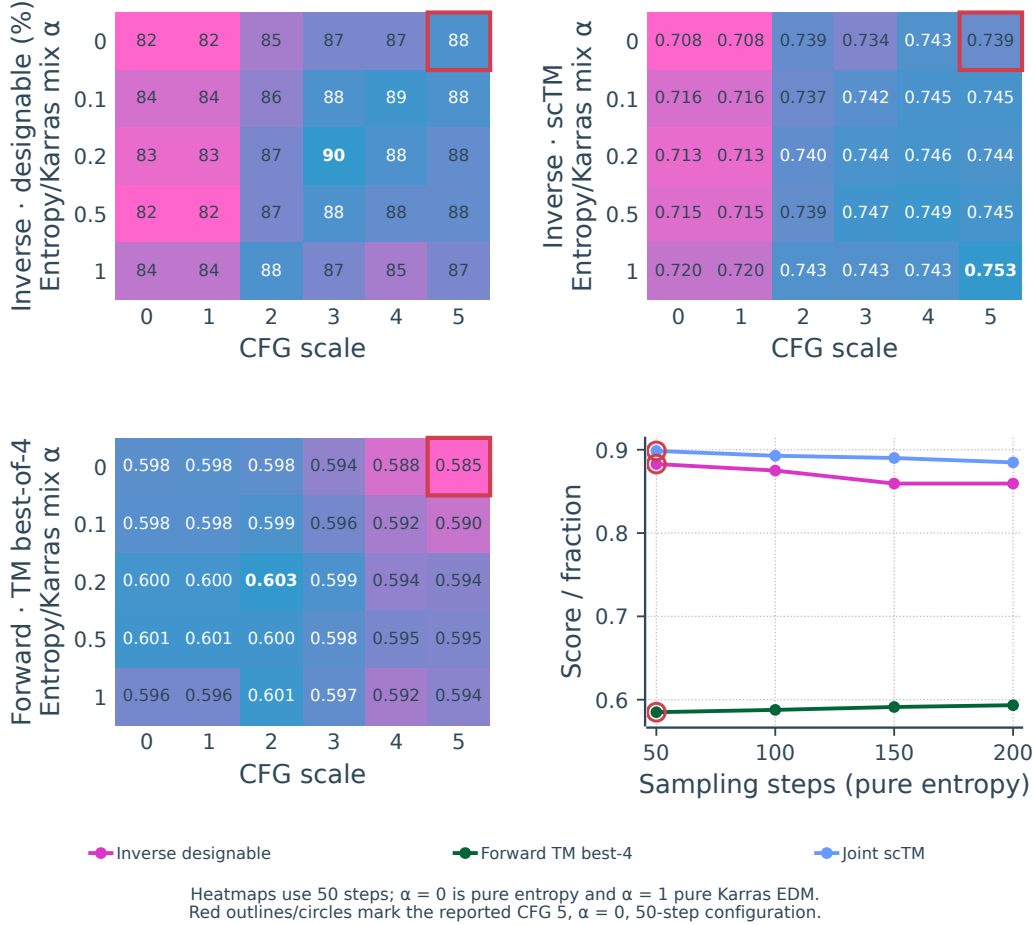


Figure 1: Sampler ablation on the frozen 396M checkpoint. Heat maps sweep CFG and entropy/Karras mixing at 50 steps; red outlines mark the locked configuration. The final panel varies the number of pure-entropic steps. No setting dominates every direction, motivating selection on inverse folding alone and reporting one configuration throughout.

operation used elsewhere in this paper. Competitive strict-threshold co-designability therefore shows that compact structure tokens need not prevent sequence–structure consistency, while the length trend exposes the present model’s limit: joint scTM is stable, but absolute coordinate error grows and forward accuracy falls for long chains. The sampler sweep likewise shows that the 50-step entropic schedule is not an isolated oracle cell (Figures 1–2). Overall, a moderately sized, from-

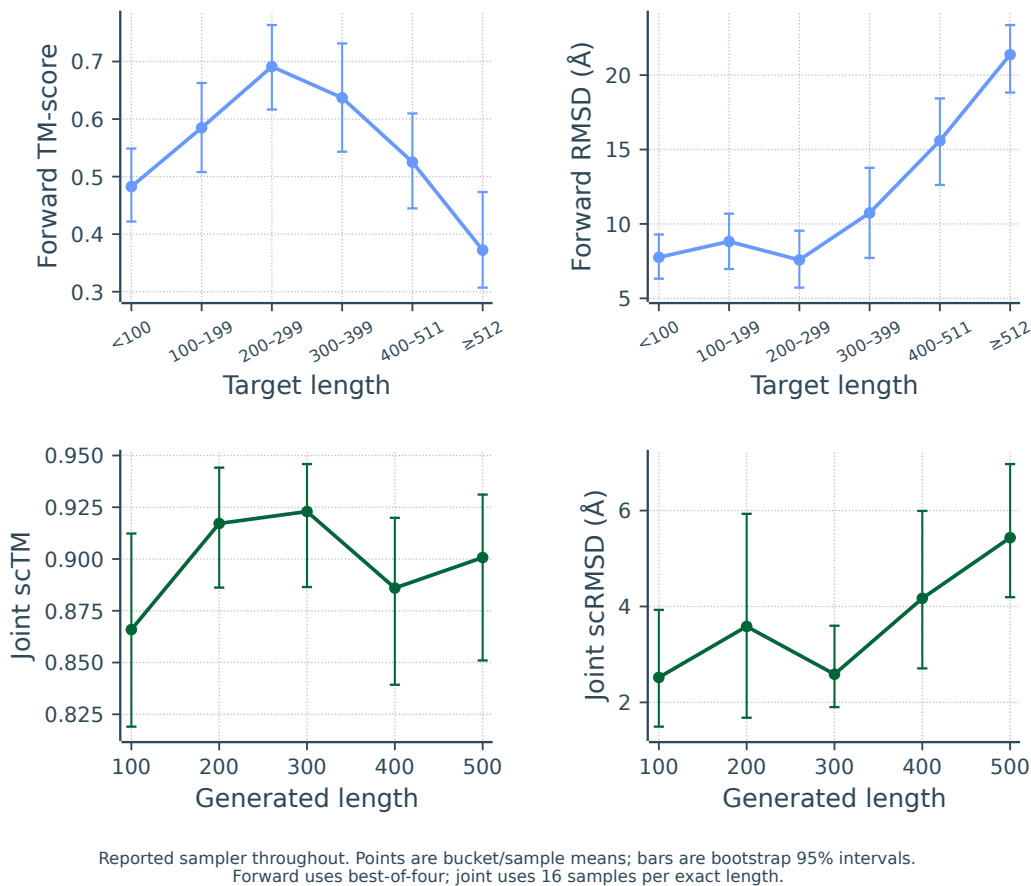


Figure 2: Length-stratified performance at the locked setting. Points are bucket/sample means and bars are bootstrap 95% intervals. Forward folding uses best-of-four samples per target; joint generation uses 16 samples at each exact length. Joint scTM remains high through 500 residues, while coordinate error rises with length and forward folding degrades most above 400 residues.

scratch CoBit checkpoint transfers the paper’s bitstream formulation to protein sequence $\leftrightarrow$ structure conditionals and unconditional co-generation with no task-specific heads or retraining.

5 DISCUSSION

[To be written.]

6 CONCLUSION

[To be written.]

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	CoBit-CC3M	CoBit-CC12M
Training corpus	CC3M (≈ 2.9 M pairs)	CC12M (≈ 11.7 M pairs)
Parameters; layers $\times$ width	462M; 24×1024	633M; 26×1152
Image tokenizer (Open-MAGViT2, 18-bit LFQ)	$f16$: $16 \times 16 = 256$	$f8$: $32 \times 32 = 1,024$
Text budget L_t ; positions; bits S	48; 310; 5,890	96; 1,126; 21,394
Training steps	1,000,000	1,744,000 (87% of target)
Relative training compute	$1 \times$	$\approx 8.7 \times$
Tokenizer rFID floor (COCO-30K)	1.592	0.726
Tokenizer PSNR (COCO-30K)	22.24 dB	26.92 dB

Table 5: The two checkpoints. CC12M has $4 \times$ the data, $1.37 \times$ the parameters, $3.63 \times$ the positions per step and $1.74 \times$ the steps ($\approx 8.7 \times$ the compute), and a materially better codec, so every image-side comparison in §A.4 is biased in CC12M’s favour. The rFID floor is the Fréchet distance between the 30,000 real images and their own tokenizer round-trip.

A IMAGE AND TEXT EXPERIMENTS

Supporting material for §4.3. Every figure was read from the job log of the run that produced it. Unless stated otherwise, sampling is deterministic DDIM on the entropy-derived grid with `carry` self-conditioning and $\sigma_{\text{dec}}=0.1$.

A.1 PROTOCOL AND CHECKPOINTS

Zero-shot MS-COCO. 30,000 captions are sampled with a fixed seed from COCO val2014 (Lin et al., 2014), one image is generated per caption, and FID (Heusel et al., 2017) is computed with Inception-V3 pool3 features against the 30,000 real images (both resized on the shorter side to 256 and centre-cropped). CLIP score is $100 \cdot \overline{\cos}(\text{image}, \text{text})$ with OpenAI ViT-B/32 (Radford et al., 2021); ViT-L/14 is given in Table 6 because published CLIP scores use that backbone and the two scales differ substantially. CMMD (Jayasumana et al., 2024) and SWD are computed in CLIP ViT-B/32 space against a 30,000-image real bank; precision and recall follow Kynkäänniemi et al. (2019) in the same space ($k=3$). Captioning uses the complete Karpathy test split (5,000 images) with `pycocoevalcap` including SPICE (Anderson et al., 2016); CLIPScore and RefCLIPScore follow Hessel et al. (2021). GenEval (Ghosh et al., 2023) uses the official harness (553 prompts $\times$ 4 images). The FID reference statistics and prompt captions are byte-identical between the $f16$ and $f8$ asset builds, so the text $\rightarrow$ image comparison between the two models is exact on the image side.

Operating points. Classifier-free guidance evaluates the conditional and null branches as one batch (§3.3), so $\text{NFE} = 2 \times \text{steps}$ for guided directions and $= \text{steps}$ for joint generation. Text $\rightarrow$ image: deterministic, guidance 2.5 (CC3M) / 2.0 (CC12M), 10 / 16 steps. Image $\rightarrow$ text CLIP: churn $\gamma=0.175$, guidance 3.0, 256 steps. Captioning: churn $\gamma=0.175$, guidance 2.0, 256 steps. Joint: deterministic, unguided. Each was chosen by a bracketed sweep at $n=5000$ read on CMMD (CIDEr for captioning) and confirmed at the full protocol where one exists.

Sample-count bias. FID is strongly biased in n : at a fixed configuration CC3M’s FID is 28.10 at $n=5000$ and 22.79 at $n=30,000$ (256 steps), and 24.00 versus 18.73 at 10 steps, so the $+5.3$ bias is a property of n alone; CMMD moves by under one point ($22.57 \rightarrow 22.16$). Sweeps are therefore read on CMMD, FID is quoted at $n=30,000$ wherever a full-protocol run exists, and FID is never compared across different n .

A.2 COMPLETE RESULTS

Table 6 extends Table 1 with SWD, the ViT-L/14 CLIP backbone, precision and recall, the alternative guidance and step settings, and the superseded 256-step text $\rightarrow$ image operating point.

A.3 SAMPLER CHARACTERISATION

Tables 7–9 give the step-count and guidance sweeps behind the operating points of §A.1.

Direction	n	gs	Steps	FID↓	CMMD↓	SWD↓	CLIP↑		Prec.	Rec.
							B/32	L/14		
<i>CoBit-CC3M</i>										
text → image	30,000	2.5	10	18.48	18.41	0.632	27.73	—	—	—
text → image	30,000	3.0	10	18.73	18.56	0.631	28.03	21.66	0.48	0.28
text → image	30,000	3.0	256	22.79	22.16	0.667	28.02	21.79	0.41	0.29
image → text	10,000	3.0	256	—	—	—	24.86	18.30	—	—
image → text	5,000	3.0	256	—	—	—	24.83	—	—	—
image → text	10,000	4.0	256	—	—	—	24.99	—	—	—
joint (unconditional)	30,000	—	10	50.19	41.53	0.904	23.10	16.87	0.36	0.22
joint (unconditional)	30,000	—	256	61.95	46.80	0.971	25.67	—	0.20	0.30
<i>CoBit-CC12M</i>										
text → image	30,000	2.0	16	16.59	17.76	0.591	26.51	—	—	—
text → image	30,000	2.0	20	16.70	17.68	0.585	26.54	—	—	—
image → text	5,000	3.0	256	—	—	—	22.98	—	—	—
image → text	5,000	4.0	256	—	—	—	23.08	—	—	—
joint (unconditional)	5,000	—	20	56.18	37.60	0.837	23.19	—	—	—
joint (unconditional)	5,000	—	256	64.47	40.33	0.878	24.01	—	—	—
<i>rFID floor, f16 / f8</i>	30,000			1.59 / 0.73	—	—	—	—	—	—

Table 6: Full version of Table 1. The CLIP column of the joint rows is pair coherence between the generated image and caption. Guidance 2.5 is marginally better than 3.0 for CC3M text→image on FID and CMMD and 0.30 worse on CLIP; L/14, precision and recall were measured at guidance 3.0 only. Moving CC3M from 256 to 10 steps raises precision from 0.41 to 0.48 with recall 0.29→0.28, so the gain is not a diversity-for-fidelity trade. On the L/14 column UniDiffuser (Bao et al., 2023) reports roughly 25–26 (text→image) and 22 (image→text) against our 21.66–21.79 and 18.30, consistent with the FID gap and its LAION-scale pre-training. Two run-to-run repeats of the 256-step headline gave FID 22.7843 and 22.7881 (CMMD 22.1704 / 22.1627), so differences above ≈ 0.05 FID at $n=30,000$ are real effects.

CC3M, text→image, gs 3.0						CC12M, text→image, gs 2.0				
Steps	NFE	FID	CMMD	SWD	CLIP	Steps	NFE	FID	CMMD	CLIP
4	8	28.55	22.71	0.729	27.45	8	16	24.97	20.90	26.29
6	12	24.79	19.13	0.662	27.86	12	24	22.21	18.20	26.51
8	16	24.28	18.81	0.653	27.99	16	32	21.68	17.84	26.56
10	20	24.00	18.76	0.647	28.04	20	40	21.96	17.79	26.54
12	24	24.03	18.95	0.647	28.05	24	48	22.22	17.84	26.57
16	32	24.68	19.40	0.647	28.05	32	64	22.81	18.21	26.60
32	64	26.13	20.85	0.661	28.03	48	96	23.58	18.90	26.58
64	128	27.21	21.59	0.671	28.00	256	512	27.22	21.25	26.45
128	256	27.79	22.18	0.676	27.96					
256	512	28.10	22.57	0.682	28.00					

Table 7: Step-count sweeps, $n=5000$. Both minima are bracketed on both sides: a shallow bowl (8–14 steps for CC3M, 16–24 for CC12M) with a steep wall below and a long monotone climb above. The gain over 256 steps is larger for CC12M (-5.54 FID / -3.46 CMMD) than for CC3M (-4.10 / -3.81); the optimum moving up with model and sequence size is the direction expected if the mechanism is error accumulation across steps, but two points do not establish a trend. CLIP is flat to within 0.1 between 8 and 256 steps for CC3M and cannot see the effect. At the full protocol ($n=30,000$, guidance 3.0) CC3M moves from FID 22.79 / CMMD 22.16 at 256 steps to 18.73 / 18.56 at 10 steps, i.e. -4.06 FID at 1/25.6 the cost.

Reading. The image optimum is about 10 steps and the text optimum about 256, a factor of 25, in a model that makes no architectural distinction between the two token populations. There is therefore no single operating point for a joint model, and any single-number report of one is a choice about which modality to favour; we quote each direction at its own optimum and say so in every table. Since architecture, weights, bitstream and σ grid are shared, the asymmetry must be a property of

Steps	CIDEr $\uparrow$	SPICE $\uparrow$	BLEU-4 $\uparrow$	METEOR $\uparrow$	ROUGE-L $\uparrow$	i2t CLIP $\uparrow$
10	0.0131	0.0088	0.0040	0.0483	0.1495	23.91
32	0.0588	0.0259	0.0139	0.0679	0.1807	24.34
64	0.0488	0.0233	0.0128	0.0665	0.1769	24.53
128	0.1304	0.0534	0.0285	0.0965	0.2192	—
256	0.1362	0.0541	0.0291	0.0975	0.2212	24.83
512	0.1032	0.0435	0.0237	0.0861	0.2019	24.90

Table 8: CC3M, text-emitting directions versus step count. Captioning is the complete Karpathy test split ($n=5000$, guidance 2.0, churn $\gamma=0.175$); i2t CLIP is the COCO reference set ($n=5000$, guidance 3.0). Captioning peaks at 256 steps and is bracketed on both sides, retaining 10% of its CIDEr at 10 steps; the 64-step point falling below 32 puts the run-to-run noise at roughly 0.01 CIDEr. Image $\rightarrow$ text CLIP rises monotonically and saturates (the last doubling buys 0.07). In the joint direction both halves come from one trajectory and move in opposite directions with the step count (CC3M, $n=5000$, 10 / 32 / 64 steps: FID 56.06 / 59.14 / 62.66, pair-CLIP 23.16 / 24.71 / 25.17); CC12M shows the same asymmetry (joint CMMD 37.60 at 20 steps versus 40.33 at 256, pair-CLIP 23.19 versus 24.01).

Guidance	CC3M			CC12M		
	FID	CMMD	CLIP	FID	CMMD	CLIP
1.0	33.65	25.80	25.32	31.29	22.32	24.80
1.5	29.71	23.10	26.51	27.80	21.39	25.82
2.0	28.61	22.44	27.19	27.22	21.25	26.45
3.0	28.10	22.57	28.00	27.46	22.32	27.10
4.0	28.06	22.71	28.44	28.53	23.24	27.42
5.0	28.19	23.27	28.72	—	—	—

Table 9: Guidance sweeps, text $\rightarrow$ image, 256 steps, $n=5000$ (noise paired across guidance values for CC3M). CMMD minimises near guidance 2, FID one to two notches higher, and CLIP rises monotonically throughout, so past the peak image quality must be judged on CMMD rather than CLIP. Both optima are bracketed on both sides. Guidance and step count are close to separable: re-sweeping at the re-tuned step counts leaves the structure unchanged (CC3M at 10 steps, guidance 2.0 / 2.5 / 3.0: CMMD 18.77 / **18.50** / 18.76; CC12M at 20 steps, guidance 1.5 / 2.0 / 2.5 / 3.0: CMMD 18.70 / **17.79** / 17.92 / 18.33), which is why the one-dimensional step sweep at fixed guidance was a valid search. CC12M loses CLIP to CC3M at every matched guidance value (by 0.52–1.02) while winning FID and CMMD at guidance 1–3.

the token statistics. A plausible account, which we have not tested, is that image tokens in an 18-bit LFQ code are locally redundant and strongly correlated across the grid, so a coarse trajectory lands in the right basin and additional steps mainly accumulate and re-inject self-conditioning error, whereas text tokens carry far more information per position, so the fine grid is doing real work. On that account the divergence should move with tokenizer bitrate; CC12M’s higher image optimum (f_8 , 1,024 positions; 16–20 versus 10 steps) is one weak data point in the predicted direction.

Other inference choices. Three further choices were measured on CC3M at 256 steps (hence far from the image optimum; these are lower bounds on image quality). *Churn* is strongly harmful for text $\rightarrow$ image—at guidance 3.0 and $n=5000$, FID 28.10 / CMMD 22.57 deterministic against 36.89 / 34.73 at $\gamma=0.1$ and 41.87 / 40.51 at $\gamma=0.175$, while CLIP moves by at most 0.06—yet the same $\gamma=0.175$ is what makes captioning work (deterministic decoding is degenerate, CIDEr 0.018; Table 12). *Schedule*: for text $\rightarrow$ image the entropy-derived and Karras grids are within noise of each other across the guidance sweep (e.g. FID 28.10 versus 27.96 at guidance 3.0); for captioning the entropy-derived grid is decisively better, CIDEr 0.136 versus 0.049 and SPICE 0.054 versus 0.023 at guidance 2.0, so a text $\rightarrow$ image-only ablation would have concluded the schedule does not matter. *Self-conditioning*: carry matches refined at guidance 3.0 (CMMD 22.30 versus 22.44, CLIP 27.98 versus 27.99) at half the NFE, so carry is used throughout. The flat CLIP columns of Tables 7 and 9 and the churn result are three independent demonstrations that CLIP score cannot arbitrate image quality for this model.

Text→image, $n=30,000$	CC3M	CC12M	Δ
operating point	10 steps, gs 2.5	16 steps, gs 2.0	
FID (raw) ↓	18.48	16.59	−1.89
tokenizer floor (rFID)	1.59	0.73	−0.87
FID above own floor	16.89	15.87	−1.02
CMMD ↓	18.41	17.76	−0.65
SWD ↓	0.632	0.591	−0.041
CLIP B/32 ↑	27.73	26.51	−1.22 (<i>worse</i>)
<i>Text-emitting directions, matched 256 steps, $n=5000$</i>			
captioning CIDEr ↑	0.1362	0.0571	2.4× <i>worse</i>
captioning SPICE ↑	0.0541	0.0347	1.6× <i>worse</i>
captioning BLEU-4 ↑	0.0291	0.0120	2.4× <i>worse</i>
captioning METEOR ↑	0.0975	0.0752	1.3× <i>worse</i>
i2t CLIP B/32, gs 3.0 ↑	24.83	22.98	−1.85
i2t CLIP B/32, gs 4.0 ↑	24.97	23.08	−1.89

Table 10: Scaling verdict at each model’s own optimum. On the image side CC12M wins every distributional metric and the margin survives floor correction: subtracting each tokenizer’s own rFID leaves CC12M 1.02 ahead, so the codec explains less than half of the raw FID gap (Fréchet distances are not additive, so the subtraction is a reporting heuristic showing the order of the confound). No reconstruction-CMMD floor has been measured, so the CMMD margin retains an unquantified codec advantage. On the text side CC3M wins every metric by margins far outside the ~ 0.01 CIDEr run-to-run noise. The image-side gap widens as the sampler approaches its optimum: at a matched 32 steps and guidance 2.0 ($n=5000$) it is 3.27 FID (26.09 versus 22.81) against 1.39 at 256 steps, so the earlier 256-step comparison was masking CC12M’s advantage.

A.4 CC3M VERSUS CC12M

Interpretation, and what it rests on. Scaling the corpus $4\times$, with the other advantages of Table 5, buys 1.89 FID on text→image (1.02 after floor correction) and costs about 60% of captioning CIDEr. Undertraining (CC12M stopped at 87% of its budget with a $3.6\times$ longer sequence) predicts a uniform deficit; noisier image–text correspondence in CC12M’s alt-text predicts the split observed, since more images improve the marginal image distribution while weaker caption–image alignment degrades everything conditional on text. The measured pattern is the second. This is an inference from a pattern across metrics, not a direct measurement of caption quality: the decisive test, a CMMD-versus-checkpoint curve over CC12M’s saved checkpoints, has not been run, and CC12M has not been evaluated on GenEval.

A.5 GENEVAL BREAKDOWN

A.6 CAPTIONING

No published zero-shot baseline exists for this number. UniDiffuser (Bao et al., 2023) evaluates image→text with CLIP score only, and L-Verse’s captioning numbers (Kim et al., 2022) are for a model trained on the MS-COCO Karpathy splits. For scale, on the $\times 100$ convention those papers use, L-Verse reaches BLEU-4 39.9 / METEOR 31.4 / ROUGE 60.4 / CIDEr 102.2 / SPICE 23.3, M^2 Transformer (Cornia et al., 2020) CIDEr 131.2 and OSCAR_L (Li et al., 2020) 140.0, all trained on MS-COCO; our 0.136 is roughly $7.5\times$ below L-Verse, not $100\times$. L-Verse’s own no-truncation variant moves CIDEr from 102.2 to 181.6 while BLEU-4 falls from 39.9 to 27.6 on the same model and images, so caption length and phrasing alone swing these metrics by tens of points: they track surface form at least as much as content. That is the failure here. The model writes fluent CC3M-style alt-text (“*example of a classic bathroom design with stainless countertops*”) against COCO references written as “*A bathroom with a shower and a sink.*”; CIDEr’s IDF-weighted n -grams penalise the register mismatch almost regardless of content, whereas on in-domain CC3M captions the same model reaches 0.90 of the real-caption CLIP ceiling. We report CIDEr for completeness and read the result through RefCLIPScore.

Model	Params	Single	Two	Count	Colors	Pos.	Attr.	Overall
Show-o (Xie et al., 2024)	1.3B	0.98	0.80	0.66	0.84	0.31	0.50	0.68
SD3	2B	0.98	0.74	0.63	0.67	0.34	0.36	0.62
SD-XL	—	0.98	0.74	0.39	0.85	0.15	0.23	0.55
SD v2.1	—	0.98	0.51	0.44	0.85	0.07	0.17	0.50
SD v1.5	—	0.97	0.38	0.35	0.76	0.04	0.06	0.43
CLIP retrieval	—	0.89	0.22	0.37	0.62	0.03	0.00	0.35
LlamaGen	0.8B	0.71	0.34	0.21	0.58	0.07	0.04	0.32
CoBit-CC3M (ours)	462M	0.80	0.12	0.05	0.46	0.04	0.03	0.249
minDALL·E	—	0.73	0.11	0.12	0.37	0.02	0.01	0.23

Table 11: Per-task GenEval scores; prior work as printed in Ghosh et al. (2023) (Table 2) and Xie et al. (2024) (Table 3). Our run used the entropy-derived grid, deterministic sampling and guidance 3.0 at 256 steps, $25\times$ past the image optimum, and has not been re-measured; the score is a lower bound. Single objects render 80% of the time and colours 46%, while compositional binding stays near the floor (attribute 0.03, position 0.04), as it does for every sub-billion-parameter model in the table; CC3M alt-text rarely specifies spatial relations or colour-object bindings. The sampler configuration matters materially: an earlier pass of the same checkpoint that had silently fallen back to a Karras grid at guidance 2.0 scored 0.209.

Configuration	BLEU-4	METEOR	ROUGE-L	CIDEr	SPICE	CLIPScore	RefCLIPScore	CLIP L/14
<i>CC3M, 256 steps</i>								
deterministic, gs 1.0 (<i>degenerate</i>)	0.004	0.054	0.157	0.018	0.012	—	—	—
entropic, churn $\gamma=0.175$, gs 4.0	0.0147	0.0695	0.1804	0.0520	0.0247	0.623	0.6678	18.34
entropic, churn $\gamma=0.175$, gs 2.0	0.0291	0.0975	0.2212	0.1362	0.0541	0.612	0.6606	17.86
Karras, churn $\gamma=0.175$, gs 2.0	0.0131	0.0672	0.1792	0.0485	0.0232	—	0.6578	—
<i>CC12M, 256 steps</i>								
entropic, churn $\gamma=0.175$, gs 2.0	0.0120	0.0752	0.1719	0.0571	0.0347	—	—	—

Table 12: Zero-shot captioning on the complete Karpathy test split ($n=5000$, no empty captions). Guidance 2.0 maximises the n -gram metrics and guidance 4.0 the CLIP-space metrics; the two families disagree, so both are reported. The entropy-derived grid beats Karras by $2.81\times$ on CIDEr while RefCLIPScore moves by 0.003.

A.7 NOTES ON COMPARABILITY AND MEASUREMENT HYGIENE

Literature figures. Every number in Table 2 was read from the cited paper’s own table or from the compilation tables of Xie et al. (2024), Bao et al. (2023) and Zhou et al. (2022). Improved VQ-Diffusion (Tang et al., 2022), our closest architectural relative (discrete diffusion over VQ tokens, trained on CC3M+CC12M), reports FID 8.44 on MS-COCO and 15.58 on Conceptual Captions; neither is a zero-shot COCO number (its COCO table compares against COCO-trained models, and the CC figure is in-domain), so it is excluded. Only the ViT-L/14 column of Table 6 may be placed beside published CLIP scores; the B/32 and L/14 scales differ substantially (28.02 versus 21.79 for the same samples). Generations faithfully reproduce CC3M’s distribution (stock-photo watermarks, white-background product shots, single subjects) against COCO’s clean multi-object scenes; that distance penalises FID independently of image quality, and Xie et al. (2024) make the same observation from the other side for models fine-tuned on aesthetic data.

Schedule audit. If the entropy tables are absent from a run directory the sampler silently reverts to a Karras grid. Every run reported here was audited for the corresponding warning; all headline, sweep and CC12M numbers come from runs with a count of zero, and the only runs in which it occurred supply the “Karras” arms of the schedule comparison, where the fallback is the treatment. The audit is not optional: the unaudited GenEval pass that scored 0.209 had fallen back.

B AUDIO AND TEXT EXPERIMENTS

[George: additional results and experimental details.]

The task is evaluating the relevance and likelihood of the predicted text continuation, given the text prompt.

You should also consider whether the meaning of the text continuation is making sense. The text prompt is: "prompt", and the text continuation is: "content"

You must give an overall rating from 1 to 5. The rating guideline is as below:

- 1: The text continuation is very unlikely and irrelevant to the text prompt.
- 2: The text continuation is unlikely and marginally relevant to the text prompt.
- 3: The text continuation is moderately likely and relevant to the text prompt.
- 4: The text continuation is likely and relevant to the text.
- 5: The text continuation is very likely and highly relevant.

You should take the following steps to provide the score:

First: briefly analyse the sample with the above definition.

Second: MUST follow the output format as: I would rate the score as _

Figure 3: LLM-as-a-Judge system message. The prompt and content placeholders are replaced with the relevant transcriptions

B.1 EXPERIMENTAL DETAILS

B.1.1 TRAINING WITH GROUND-TRUTH SPEAKER FEATURES

For all three conditional tasks, BiCodec-global speaker tokens are part of the condition. For ASR, speaker tokens are extracted from the full speech signal, which is always provided as input. As already mentioned in Subsection 4.4, inference-time speaker tokens are extracted from the audio prompt for continuation and from a held-out recording for TTS to rule out information leakage. During training, however, speaker features from the full ground-truth audio samples are used for all tasks. This forces CoBit to learn a tighter relationship between speaker features and speech tokens, simultaneously accounting for the variability in BiCodec-global, which does not guarantee that all recordings of the same speaker produce identical outputs. A training-inference mismatch is therefore introduced to facilitate unified training.

B.1.2 EVALUATION PROTOCOL

CoBit is evaluated on four text-audio tasks: co-generation, speech continuation, TTS and ASR. The goal is to generate consistent streams of text and audio with fluent, intelligible and natural content. To assess the linguistic properties of generated speech signals, we transcribe them using an external whisper-large (Radford et al., 2023) ASR model.

The alignment between two text sequences is quantified as the percentage of mismatches at the word (WER) or character (CER) level. For co-generation, error rates are computed between the two generated sequences; for TTS, between reference text and the transcriptions of the generated audio; for ASR, between reference and generated text. The fluency of a text sequence is measured by its perplexity under an external language model (in this case Llama-3.2 1B). The metric is denoted GenPPL and is computed separately for each modality in co-generation and for speech transcriptions in speech continuation. To further determine the conceptual relevance of produced continuations, GPT-4o is employed in an LLM-as-a-Judge setting, assigning each continuation a score from 1 to 5. The system message is taken directly from Tseng et al. (2025) and presented in Figure 3.

The most reliable evaluation of generated audio is human perception, which we approximate by UT-MOS (Saeki et al., 2022), a neural mean opinion score (MOS) predictor that evaluates naturalness on a 1-5 scale (1: bad, 2: poor, 3: fair, 4: good, 5: excellent). In zero-shot TTS, however, natural-sounding audio is insufficient; the goal is to also replicate the voice in the original utterance. This is captured by speaker similarity (SpkSim), which is computed as the cosine similarity between the outputs of an ECAPA-TDNN speaker verification model. For speech continuation, we are also interested in how closely the generated audio follows the distribution of ground-truth continuations. We thus measure Fréchet Speech Distance (FSD), an adaptation of the FID metric to audio. Population statistics are extracted from self-supervised representations of the true and generated continuations, and their Fréchet distance is calculated. Following previous work (Chou et al., 2025), two variants are considered: FSD-wlm, which is based on embeddings from the 6th layer of WavLM-large (Chen et al., 2022) to measure frame-level match, and FSD-e2v, which is based on expressiveness representations from Emotion2Vec (Ma et al., 2024). FSD metrics do not evaluate the quality of the generated audio; they capture its population-level resemblance to the original continuations.

The SALMON benchmark is used for acoustic evaluation. In its original form, SALMON creates replicas of short speech recordings where the start remains identical, but some paralinguistic property (sentiment, speaker identity/gender, background, room) is modified midway. SLMs receive an accuracy score related to how often the positive example is assigned higher likelihood than the negative one. Given the incompatibility of likelihood-based evaluation for diffusion models previously covered in Subsection 4.4, we turn to an alternative generation-based SALMON formulation (Hsu et al., 2026). The examined SLM produces a continuation of the shared start of the two examples, and a judge model calculates embeddings for the positive, negative and generated continuations. The SLM then receives an accuracy score based on how often the generated continuation is closer to the positive example than the negative one, in terms of cosine similarity in the embedding space. TITANET (Koluguri et al., 2022) is used as the judge for sentiment, speaker and gender tasks; HuBERT-large-audioset and Wav2Vec2-large-audioset are used for background and room tasks, respectively.

Experimental directions that heavily rely on information that SLMs most commonly learn through text initialization, such as spoken question answering or dialogue-based evaluation, were omitted from this work, as CoBit is initialized from scratch and is considerably smaller than state-of-the-art dialogue systems.

B.1.3 BASELINES

The most comparable text-audio baselines for CoBit are models that learn to generate both text and speech. TASLM (Tseng et al., 2025) generates the two modalities jointly, leveraging continuous text-aligned speech representations from the TASTE encoder and implementing separate prediction heads. The semantic equivalence of text units and speech embeddings bridges the gap between audio and text generation, producing linguistically fluent and conceptually consistent audio. Other approaches focus on systems that can be extended to cross-modal tasks. SpiritLM (Nguyen et al., 2025) is trained on interleaved text-audio sequences, and its expressive variant for additional control over pitch and style. SpiritLM models can perform TTS and ASR via task-specific fine-tuning and few-shot prompting. Moshi (Défossez et al., 2024) is a spoken dialogue system that produces text alongside speech, as an inner monologue that steers generation towards intelligible sentences. Its cross-modal capabilities arise from introducing a time delay on the generated stream and inference-time teacher-forcing on the conditioning stream. GPA (Cai et al., 2026) is a multi-task autoregressive hybrid that treats TTS and ASR as conditional language modelling problems. It is the closest method to CoBit in terms of scale, multi-task joint training and initialization from scratch, but it is limited to cross-conditional tasks and cannot perform speech language modelling. However, all of these models require task-specific design and modifications for TTS and ASR, in contrast to CoBit.

In the context of speech generation, additional comparisons can be drawn to textless SLMs, which mainly rely on the powerful Mimi tokenizer to computationally balance the semantic benefits of text generation with fine-grained acoustic details. Llama-Mimi (Sugiura et al., 2025) autoregressively predicts each Mimi layer, leveraging RVQ’s conditional nature: the generation of each code is conditioned on both the previous tokens and the earlier codebooks of the same token. Flow-SLM (Chou et al., 2025) exploits Mimi’s semantic-acoustic combination for cascaded prediction, breaking the strictly autoregressive paradigm: a causal transformer predicts the semantically distilled first

Model	Parameter Count	Modalities	Text Initialized	Speaker-Conditioned Decoder	Method
Flow-SLM-S	395M	Speech-only	✓	✗	Semi-Autoregressive (flow matching)
Flow-SLM-L	1.25B				
Flow-SLM-Ext.	1.25B				
SpiritLM-Base	7B	Speech+text	✓	✓	Word-level interleaving
SpiritLM-Exp.					
Llama-Mimi	1.3B	Speech-only	✓	✗	Flattened RVQ
TASLM-Embed	1.3B	Speech+text	✓	✓	Speech-text alignment
GPA-v1.5	505M	Speech+text	✗	✓	Autoregressive TTS-ASR Hybrid
Cascade	640M	Speech+text	✓	✓	Text LM + Speaker Sampler + TTS

Table 13: Overview of text-audio baselines. Parameter counts refer to the generative models themselves and exclude the encoders and decoders. The only difference between Flow-SLM-L and Flow-SLM-Ext. is that the latter is trained on an augmented audio corpus.

RVQ layer; then, unquantized Mimi embeddings are inferred with flow matching, conditional on look-ahead semantic tokens.

None of the multimodal systems are evaluated on unconditional co-generation. TASLM-Embed and Moshi are the only baselines that are, in theory, capable of true parallel generation, but they are not trained for such a task. Instead, we compare CoBit to a cascaded pipeline: text is generated unconditionally with a text LM and speaker identity is sampled from a small transformer; a TTS system then uses the text and speaker features to synthesize the speech sequence. This cascade system effectively targets the same distribution as CoBit for joint generation, factorized as $p(\text{text}, \text{speaker}, \text{speech}) = p(\text{text})p(\text{speaker})p(\text{speech}|\text{text}, \text{speaker})$. To match the 630M parameter scale of CoBit, `SmolLM2-135M` (Allal et al., 2025) is used for text generation and Spark-TTS (Wang et al., 2025) is used for speech synthesis. Speaker features are sampled by a transformer with 1M parameters in the form of BiCodec-global tokens and passed to the Spark-TTS decoder. We train the speaker transformer on LibriTTS and MLS-English. All three components of the cascade are autoregressive. A known drawback is that unconditional generation with `SmolLM2-135M` can lead to web or code artifacts that cannot be rendered into natural speech, causing very high GenPPL-Speech, as observed in Table 14.

The different baselines are summarized in Table 13. Unless marked with an asterisk (*), all baselines are evaluated internally using publicly available checkpoints.

B.2 ADDITIONAL RESULTS

Tables 14, 15, 16 and 17 report the full evaluation for co-generation, continuation, SALMON and cross-modal tasks, respectively.

Samplers with 256 and 512 neural function evaluations (NFE) are considered separately, with stochasticity $\gamma \in \{0.11, 0.13, 0.15, 0.17, 0.19, 0.22, 0.25, 0.28\}$ for $\text{NFE} = 256$ and $\gamma \in \{0.15, 0.19, 0.23, 0.27, 0.31, 0.34, 0.37, 0.4\}$ for $\text{NFE} = 512$. In all cases, guidance scales $\omega = 5$ and $\omega = 6$ were examined. Higher stochasticity values were preferred for $\text{NFE} = 512$ in comparison to $\text{NFE} = 256$. With a few exceptions, increasing the number of function evaluation does not improve performance; this is likely due to the softer peak of the learned entropy-rate profile.

C PROTEIN SEQUENCE AND STRUCTURE EXPERIMENTS

Model	WER (%) ↓	CER (%) ↓	GenPPL-Text ↓	GenPPL-Speech ↓	UTMOS ↑
Text-LM+Speaker+TTS Cascade	3.2	3.1	122	1,648	4.23
CoBit					
NFE = 256	5.7	3.6	240.2	145.3	3.73
NFE = 512	3.7	2.3	280	141.2	3.78

Table 14: Joint generation performance, computed by generating 1,024 samples. Speech perplexity for the cascade system is high, as SmolLM produces multilingual sequences or web/code artifacts that are not transferable to speech. Filtering out these rows substantially improves GenPPL-speech.

Model	UTMOS ↑	FSD-wlm ↓	FSD-e2v ↓	GenPPL ↓	GPT-4o ↑
Autoregressive SLMs					
Flow-SLM-S	2.47	3,391.4	35.7	242.8	1.1
Flow-SLM-L	3.48	3,476.4	35.4	131.7	1.57
Flow-SLM-Ext.	3.47	3,445.1	33	122.4	1.67
SpiritLM-Base	3.2	7,325.4	26.8	101.4	2
SpiritLM-Exp.	3.17	6,293.6	16.9	153.2	1.62
Llama-Mimi	3.18	3,621	14.1	98.5	1.58
TASLM-Embed	4.16	1,410.3	9.9	38.6	3.42
CoBit					
NFE = 256	3.9	2,183.7	11.9	85.2	1.44
NFE = 512	3.89	2,386.3	13	83.6	1.45

Table 15: Speech continuation performance on LibriSpeech-PC test-clean. GPT-4o and GenPPL are computed on the transcription of the generated speech. For GenPPL, the transcriptions of the prompt and the generated continuation are concatenated. The prompt audio is fixed to the first 3 seconds of each test sample.

Model	Sentiment	Speaker	Gender	Bg.- Domain	Bg.- Random	Room
Autoregressive SLMs						
Flow-SLM-S*	85	87	92.5	77	86	69.5
Flow-SLM-L*	92.5	99.5	99	77	84.5	77
Flow-SLM-Ext.*	93	98	100	77	86.5	85.5
SpiritLM-Base*	47.5	52	50.5	50.5	52.5	50
SpiritLM-Exp.*	72	54	56	58.5	55.5	54.5
Llama-Mimi*	93	90	95	81	90	78.5
TASLM-Embed*	95	100	100	61.5	54.5	57.5
CoBit						
NFE = 256	90	94.5	97	69	81	74
NFE = 512	90.5	93	93.5	73.5	80	75.5

Table 16: Speech continuation performance across the consistency tasks of the SALMON benchmark for generation-based evaluation. Each model generates a continuation for each prompt, and accuracy scores are computed for each task by comparing the cosine similarity of the generated continuation with the positive and negative examples. Different embedding models are used for each task, specified in Appendix B.1.3.

Model	Text-to-Speech				Speech-to-Text	
	WER% ↓	CER% ↓	SpkSim ↑	UTMOS ↑	WER% ↓ Clean	WER% ↓ Other
Autoregressive SLMs/Hybrids						
SpiritLM-Base*	-	6.7	-	-	6	11
Moshi*	4.7	-	-	-	5.7	-
GPA-v1.5	3.1	3.1	0.4	4.37	3.4	6.3
CoBit						
NFE = 256	5.3	4.9	0.27	4.03	8.1	22.4
NFE = 512	5.3	4.1	0.27	4	8.2	22

Table 17: Text-to-speech and Speech-to-text performance on LibriSpeech-PC. For text-to-speech, evaluation is conducted on the test-clean partition. For each speaker in the dataset, one sample was held out to extract speaker features. Speech-to-text evaluation is conducted on test-clean and test-other partitions separately.

[Bruno: additional results and experimental details for reproducibility.]